Econometrics — Exam-Eve Summary (Key Points Only)

Wooldridge Ch.1–6 · Econometrics 1–2 · course R code | **[S]** not in material, standard form · **[?]** needs check · **[D]** derived from material · ✔ recomputed in code

Ch.1 · Nature of Econometrics & Data

1.1 Core definitions

- Econometrics = statistical methods to analyze economic data | mostly **nonexperimental** data
- Goals: estimate relationships | test theories | evaluate policy
- Most of econometrics = specification of the error u
- **Ceteris paribus** = "other relevant factors being equal" | causal questions hold other factors fixed
- Population = entire set to study | sample = subset | **external validity** = conclusions valid outside the study

1.2 Data types (→ method choice)

Type	Key feature
Cross-sectional	one point in time | ≈ independent (random sampling)
Time series	serially correlated | order matters | trends, seasonality
Pooled cross sections	independent samples in different years | policy before/after
Panel	**same units** over time | controls time-invariant unobservables | lags

Ch.2 · Simple Regression

Slide equations are images (not extracted) → formulas are **[S]** unless marked.

2.1 Model & OLS

$y = \beta_0 + \beta_1 x + u$ | $E(u|x) = 0$ → PRF $E(y|x) = \beta_0 + \beta_1 x$ **[S]**

$\hat{\beta}_1 = \Sigma(x_i - \bar{x})(y_i - \bar{y}) / \Sigma(x_i - \bar{x})^2$ (num = covariation, den = $SST_x > 0$)
$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$ **[S]**

- Procedure: $\bar{x}$, $\bar{y}$ → deviations → Σ cross products → Σ(x dev)² → $\hat{\beta}_1$ → $\hat{\beta}_0$
- Residual $\hat{u}_i = y_i - \hat{y}_i$ (actual − fitted) | $\Sigma\hat{u}_i = 0$ | $\Sigma x_i\hat{u}_i = 0$ | line through $(\bar{x}, \bar{y})$ **[S]**

2.2 CEO salary example (slide table)

- Fitted line from table: **salaryhat = 963.191 + 18.501·roe** **[D]**
- Slope = (2004.808 − 1072.348)/(56.3 − 5.9) = **18.501** **[D]**
- Salary in \$1,000s → CEO 12: 933 − 1459.023 = **−526.023** → over-predicted by \$526,023
- CEO 5: 1368 − 1218.508 = **+149.493** → under-predicted
- Slide typo: CEO 7 residual = 1078 − 1266.611 = **−188.611** (slide: +) ✔

2.3 Goodness of fit

$SST = \Sigma(y_i - \bar{y})^2$ | $SSE = \Sigma(\hat{y}_i - \bar{y})^2$ | $SSR = \Sigma\hat{u}_i^2$
$SST = SSE + SSR$ → $R^2 = SSE/SST = 1 - SSR/SST$ **[S]**

- Simple regression: $R^2 = corr(y, x)^2$ ✔ (regression.R)

2.4 Log forms (interpretation)

Model	β_1 means
y on x	$\Delta y = \beta_1 \Delta x$
y on log x	$\Delta y \approx (\beta_1/100)\cdot\%\Delta x$ **[S]**
log y on x	$\%\Delta y \approx (100\cdot\beta_1)\Delta x$ → **semi-elasticity**
log y on log x	$\%\Delta y \approx \beta_1\cdot\%\Delta x$ → **constant elasticity**

2.5 Assumptions & theorems

- SLR.1 linear in parameters | SLR.2 random sampling | SLR.3 sample variation in x | SLR.4 $E(u|x)=0$ | SLR.5 $Var(u|x)=\sigma^2$
- **Thm 2.1** (SLR.1–4): $E(\hat{\beta}_j) = \beta_j$ → unbiased = correct **on average** over repeated samples

Thm 2.2 (SLR.1–5): $Var(\hat{\beta}_1) = \sigma^2/SST_x$ **[S]**
Thm 2.3: $\hat{\sigma}^2 = SSR/(n-2)$ | $se(\hat{\beta}_1) = \hat{\sigma}/\sqrt{SST_x}$ **[S]**

- Var ↑ with σ^2 (noise) | Var ↓ with more variation in x

2.6 Binary x & treatment effects

- $x \in \{0,1\}$: β_0 = mean of y if x=0 | β_1 = **difference in means** **[S]**
- ATE $\tau = E[y_i(1) - y_i(0)]$ | never observe both outcomes | **random assignment** → OLS unbiased for τ
- Job training: +\$1,790 → 1,790/4,550 = **39.3%** ✔

Ch.3 · Multiple Regression: Estimation

3.1 Model & interpretation

$y = \beta_0 + \beta_1 x_1 + \ldots + \beta_k x_k + u$ **[S]** | "linear" = linear in **parameters**

- β_j = effect of x_j **holding other x fixed** (ceteris paribus)
- College GPA: hsGPA +1 (ACT fixed) → colGPA **+0.453**
- Econometrics 2: in multiple regression β_1 is the **partial slope**, not the slope of y on x_1
- Partialling out: regress x_1 on other x → residual $\hat{r}_1$ → regress y on $\hat{r}_1$ → slope = $\hat{\beta}_1$

3.2 Arrest example (implied coefficients [D])

- Prop. prior arrests +0.5 → −7.5 per 100 men → coef. **−0.150**
- Months in prison +12 → −0.408 per man → **−0.034**/month
- Quarters employed +1 → −10.4 per 100 men → **−0.104**
- Small R^2 can still give good ceteris paribus estimates

3.3 MLR assumptions

- MLR.1 linear | MLR.2 random sample | MLR.3 **no perfect collinearity** | MLR.4 $E(u|x_1...x_k)=0$ | MLR.5 homoskedasticity
- Unbiased: MLR.1–4 | **Gauss–Markov**: MLR.1–5 → OLS = **BLUE**

3.4 Omitted variable bias

True $y = \beta_0+\beta_1x_1+\beta_2x_2+u$, omit x_2:
$E(\tilde{\beta}_1) = \beta_1 + \beta_2\cdot\tilde{\delta}_1$, $\tilde{\delta}_1$ = slope of x_2 on x_1 [S]

Bias [S]	corr(x_1,x_2) > 0	corr < 0
$\beta_2 > 0$	**upward (+)**	downward (−)
$\beta_2 < 0$	downward (−)	**upward (+)**

- No bias if $\beta_2 = 0$ (irrelevant) or $\tilde{\delta}_1 = 0$ (uncorrelated)
- Ability omitted in wage eq. → return to educ biased upward [S]
- Irrelevant variable included → no bias, variance may ↑

3.5 Variance & multicollinearity

$Var(\hat{\beta}_j) = \sigma^2 / [SST_j(1 - R_j^2)]$, R_j^2 = R^2 of x_j on other x
$\hat{\sigma}^2 = SSR/(n - k - 1)$ | $se(\hat{\beta}_j) = \hat{\sigma}/\sqrt{[SST_j(1-R_j^2)]}$ | VIF = $1/(1-R_j^2)$ [S]

- σ^2 ↑ → Var ↑ | SST_j ↑ → Var ↓ | R_j^2 ↑ → Var ↑ | R_j^2 → 1 → Var → ∞
- Multicollinearity inflates only the involved variables' variances | **not** an MLR.3 violation
- Dropping a variable reduces variance but may cause OVB → bias–variance trade-off

Ch.4 · Inference

4.1 Distribution

- MLR.6: u ~ Normal(0, σ^2), independent of x | MLR.1–6 = CLM [S]
- Normality impossible: wages (≥ 0), arrests (few integers), unemployment (0/1) | log(y) may help

$t = (\hat{\beta}_j - a_j) / se(\hat{\beta}_j) \sim t_{(n-k-1)}$ (a_j = 0 usually) [S]

4.2 Critical values & rules

H_1	Reject if	Slide (5%)
$\beta_j > 0$	t > c	df 28: **1.701** ✓
$\beta_j < 0$	t < −c	df 18: **−1.734** ✓
$\beta_j \neq 0$	|t| > c	**2.06** (df ≈ 25 [?])

- Large df (two-sided): 10% **1.645** | 5% **1.96** | 1% **2.576** [S] ✓
- Steps: H_0/H_1 → t → df = n−k−1 → c (α, one/two-sided) → decide → interpret

4.3 p-value

- p = smallest significance level at which H_0 is rejected | reject ⇔ **p < α**
- Two-sided: P(|t| > 1.85) = 2 × 0.0359 = **0.0718** ✓

- Statistical ≠ economic significance → always discuss magnitude | "wrong" sign → misspecification?

4.4 Confidence interval

$\hat{\beta}_j \pm c \cdot se(\hat{\beta}_j)$, c = 97.5th pct of $t_{(n-k-1)}$ (95%) [S]

- Random bounds: covers β_j in 95% of repeated samples
- a_j outside 95% CI ⇔ reject H_0: $\beta_j = a_j$ at 5% (two-sided)

4.5 Linear combination ($\beta_1 = \beta_2$)

- $t = (\hat{\beta}_1 - \hat{\beta}_2)/se(\hat{\beta}_1 - \hat{\beta}_2)$ | se needs Cov → not in standard output [S]
- Trick: $\theta = \beta_1 - \beta_2$ → regress y on x_1 and $(x_1 + x_2)$ → se($\hat{\theta}$) directly [S]

4.6 F test

$F = [(SSR_r - SSR_{ur})/q] / [SSR_{ur}/(n-k-1)] \sim F(q, n-k-1)$
R^2 form: $F = [(R^2_{ur} - R^2_r)/q] / [(1-R^2_{ur})/(n-k-1)]$
Overall: $F = (R^2/k) / [(1-R^2)/(n-k-1)]$ [S]

- q = number of restrictions | F ≥ 0 | reject if F > c
- regression.R: (0.07449/2)/((1−0.07449)/82) = **3.30** ✓
- Restricted model with different y (e.g. rational house assessments) → use **SSR form** only [S]

Ch.5 · OLS Asymptotics

Property	Needs
Unbiased	MLR.1–4
Variance formulas, BLUE	MLR.1–5
Exact t / F	MLR.1–6
Consistent	MLR.1–4 (weaker MLR.4′ suffices)
Asymptotic t / F / CI	MLR.1–5 (no normality)

- Consistency: P(estimate close to true value) → 1 as n ↑ | minimum requirement

- MLR.4′: $E(u) = 0$, $Cov(x_j, u) = 0$ [S]
- plim $\hat{\beta}_1 = \beta_1 + Cov(x_1,u)/Var(x_1)$ | inconsistency doesn't vanish with n [S]
- Large n: t ≈ N(0,1) → t, F, CI valid **without MLR.6** | homoskedasticity still needed
- $se(\hat{\beta}_j) \approx c_j/\sqrt{n}$ → why large samples are better [S]

Ch.6 · Further Issues

6.1 Logs

- Pros: % / elasticity reading | invariant to rescaling | less outliers | helps normality, homoskedasticity
- Don't log: years | percentage points | zero or negative values

6.2 Quadratic

$y = \beta_0 + \beta_1x + \beta_2x^2 + u$ → **$\Delta\hat{y} \approx (\hat{\beta}_1 + 2\hat{\beta}_2x)\Delta x$**
Turnaround **$x^* = |\hat{\beta}_1 / (2\hat{\beta}_2)|$** [S]

- Wage peaks at **24.4 years** of experience (coefficients in image [?])

6.3 Interaction & APE

$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + u \rightarrow \partial y/\partial x_2 = \beta_2 + \beta_3 x_1$ [S]

- β_2 alone = effect at $x_1 = 0$ | center ($x_1 - \mu_1$) → coefficient = effect at the mean
- APE = average of each unit's partial effect over the sample

6.4 Adjusted R^2

$R^2 = 1 - (1 - R^2)(n - 1)/(n - k - 1)$ [S]

- regression.R: $1 - (0.92551)(84/82) =$ **0.05192** ✓
- $R^2 \uparrow \Leftrightarrow$ new regressor's $|t| > 1$ | use for nonnested models
- Never compare R^2 / $\bar{R}^2$ across **different dependent variables**
- High $R^2 \neq$ causal | low $R^2 \neq$ bad partial effects

6.5 Too many controls · prediction

- Don't control for mechanisms: beer consumption (beer tax → fatalities) | doctor visits (pesticides → health spending)
- Add regressors uncorrelated with x → $\sigma^2 \downarrow$ → more precise
- log(y) model: exp(predicted log y) underestimates → $\hat{y} = \hat{\alpha}_0 \cdot \exp(\ldots)$, $\hat{\alpha}_0 = n^{-1}\Sigma\exp(\hat{u}_i)$ [S]
- CI for mean E(y|x) narrow | prediction interval for one new y wider ($+\hat{\sigma}^2$) [S]

Ch.R · R Code — Reading the Output

R.1 regression.R (Weidmann & Deming 2021)

- n = **85** groups | y = cryptography score | x: RMET (eye), skill, Ravens | all standardized → intercept ≈ 0
- `lm(perform ~ eye)` (outcome first) | `summary()` | `confint()` | `predict()`

Model	$\hat{\beta}$ (se)	p	R^2
~ eye	0.192 (0.108)	0.079	0.037
~ skill	0.214 (0.107)	0.049	0.046
~ ravens	0.217 (0.107)	0.046	0.047
~ skill + ravens	0.171 / 0.175 (0.110 each)	0.122 / 0.115	0.074

- All recomputed in R ✓
- Overall F(2, 82) = **3.3**, **p = 0.04184** → reject H_0: both = 0
- Each significant alone (p ≈ 0.05) but not when both included (corr 0.246, computed) → partial effects, larger se
- Skill **99%** CI [−0.069, 0.497] contains 0 while p = 0.049 → consistent (not significant at 1%)
- Simple regression: $F = t^2$ ($1.997^2 = 3.989$) ✓
- `predict(skill=0.5, ravens=1.5)` = **0.3475** | (1.5, 0.5) = **0.3441** ✓
- Trap: `eye_regr` overwritten by skill model → later CI-band plot wrong

R.2 Simulation (simple regression demonstration.R)

- True $\beta_0 = 3$, $\beta_1 = 0.2$ | n = 10 | `rnorm(n, 10, 2)` → 3rd arg = **SD**
- 10,000 samples: mean $\hat{\beta}_1$ = **0.2027 ≈ 0.2** → unbiasedness ✓
- Reject 1,908 / 10,000 (19.1%) at 5% → H_0 is false → this is **power** ✓
- Larger slope or n → power ↑ | true null → reject ≈ 5% [S]
- 20 CIs (slope 2): **19/20** contain 2 ≈ 95% coverage ✓

R.3 Lab 2 & Lab 4 essentials

- $\Sigma(x - \bar{x}) = 0$ always | `var()` = SS/(**n−1**) | `sd()` = √var
- competence: SS 124.07 → var **4.772**, sd **2.185** ✓
- IQR = Q3 − Q1: competence 6.5 − 4 = **2.5** | breaks 4.5 − 1 = **3.5** ✓
- Skewed data → IQR better than SD | boxplot outlier beyond 1.5×IQR (grade 36) ✓
- `to12("12")` → "12 AM" (noon bug) | `"13" - 12` → error (character) ✓

Exam-Eve Trap Checklist

C.1 Signs & units

- Residual = **actual − fitted** ($\hat{u} < 0$ → over-predicted)
- OVB sign = sign(β_2) × sign(corr(x_1, x_2))
- log–level: **×100** for % | log–log: β = elasticity | level–log: ÷100
- Check units (CEO salary in $1,000s)
- Quadratic $\beta_2 < 0$ → maximum at x*

C.2 Degrees of freedom

- Simple $\hat{\sigma}^2$: **n − 2** | multiple: **n − k − 1** | sample variance: **n − 1**
- F: numerator q | denominator n − k − 1

C.3 Commonly confused

- R_j^2 (x_j on other x) ≠ model R^2
- Two-sided p = 2 × one tail
- One-sided vs two-sided critical value (1.701 vs 2.06)
- Multicollinearity ≠ perfect collinearity
- Unbiased ≠ consistent | BLUE needs homoskedasticity
- R^2 form of F invalid if y differs

C.4 R pitfalls

- `set.seed()` first | reused object names overwrite
- `as.integer()` before arithmetic on strings
- `==` on doubles → `round()`

C.5 Written answers

- "Holding [other x] fixed, a one-unit increase in x is associated with a $\hat{\beta}$ [unit] change in y"
- Causal claim → cite MLR.4 / random assignment
- Test: H_0/H_1 → stat → df → c or p → decision → economic meaning
- Name the assumption (SLR/MLR number) behind each property